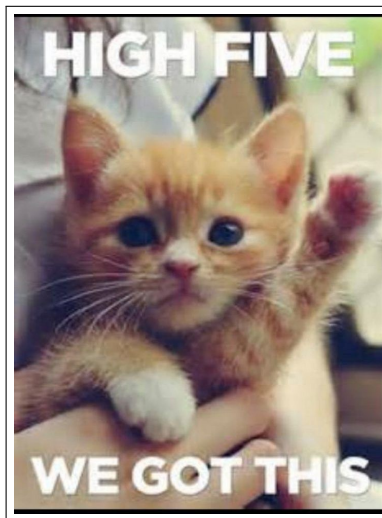


MTH310/MTH520: Graph Theory (GT)

Midsem Exam

March 2, 2025



A motivational cat that gives you emotional support for ninety minutes

Solve Q1/Q2, Q3/Q4 and Q5 in your answer booklet within ninety minutes. In case both Q1 and Q2 (or both Q3 and Q4) are attempted, the answer with the maximum marks will be counted for the final score.

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1. These are YES/NO questions based on whether these objects exist or not. If you pick YES, give an example and if you pick NO, give a short (one or two sentences should suffice) argument of why no examples exist.
- a) A self-complementary tree. [CO1]
 - b) A graph with degree sequence $(4, 4, 3, 3, 3, 3, 2, 1, 1)$. [CO2]
 - c) A graph G and an edge $e \in E(G)$ such that G and $G \setminus \{e\}$ is Eulerian. [CO1]
 - d) A graph invariant f with $f(G \cup H) = \min(f(G), f(H))$ if $V(G) \cap V(H) = \phi$. [CO2]
 - e) A graph G such that $G \square \overline{G} \cong L(G)$. [CO1]

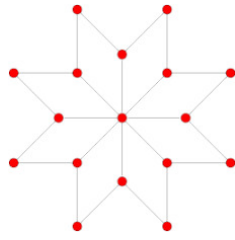
[1+1+1+1+1=5 marks]

OR

2. In Quiz 2, you have seen that the hypercube family of graphs Q_n are all median graphs. Another well-known family of median graphs are trees:
- Consider a tree T arbitrarily rooted at $r \in V(T)$. For two vertices $x, y \in V(T)$, show that the set of common ancestors L_{xy} is non-empty. [CO1]
 - Let us define the maximal element of L_{xy} as the lowest common ancestor (or LCA) of x and y . Show that $\text{LCA}(x, y)$ is unique. [CO1]
 - If the levels of x, y and $\text{LCA}(x, y)$ are a, b and c respectively in the tree order, show that $d_T(x, y) = a + b - 2c$. [CO2]
 - Now use all the above parts to conclude that every tree is a median graph. [CO4]

[1+1+1+2=5 marks]

3. Since we are in Delhi, let us look at a family of graphs associated with the Mughal empire. On the tomb of Nur-ud-din Muhammad Salim, known better by his imperial name Jahangir, lies carvings of symbols each consisting of 16 points in a loop and a central point connected to every even position point in the loop.



(a) $\mathcal{J}_{2,8}$ as described above



(b) Jahangir Watching an Elephant Fight, by Farrukh Chela

In more graph-theoretic terms, we define the Jahangir graph $\mathcal{J}_{n,m}$ as a graph with order $nm+1$ where we have an induced subgraph isomorphic to C_{nm} and an additional vertex connected to every n^{th} vertices in C_{nm} .

- Enumerate all the parameters $n, m \in \mathbb{N}$ ($0 \notin \mathbb{N}$ unlike Quiz 2) for which the following occur: [CO1]
 - $\mathcal{J}_{n,m} \cong \mathcal{J}_{m,n}$
 - $\mathcal{J}_{n,m} \cong L(\mathcal{J}_{n,m})$
 - $\mathcal{J}_{2,m}^n \cong K_{2m+1}$, where $m \geq 4$.

(Hint: Note that isomorphism implies number of vertices and number of edges of both graphs should match. But beware, the numbers matching does not imply the existence of an isomorphism.)

[2+2+2=6 marks]

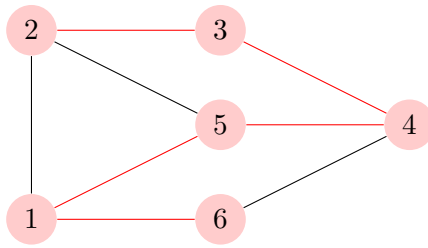
- The case when $n = 1$ (also known as wheel graph W_n) is just $K_1 + C_{nm}$.
 - Show that $W_n = \mathcal{J}_{1,n}$ contains a cycle of length k , where $3 \leq k \leq n+1$. [CO4]

- ii) Prove that when n is at least 3, W_n can be partitioned into 3 independent sets if n is even and 4 independent sets if n is odd. [CO4]

[2+2=4 marks]

OR

4. As part of a government-sanctioned development scheme in rural Dumka, Jharkhand, the president of the Zila Parishad, Birendra plans to connect all the villages in the district via newly made roads. However, the local oligarch Bablu owns all the buses in the district and is known for instructing his bus drivers to always take the longest possible path between any two villages, in order to overcharge his passengers.



(a) $D_G(2, 6) = 5 \neq 2 = d_G(2, 6)$



(b) Jil Pitha, a local delicacy in Dumka

In more graph-theoretic terms, the definition of graph distance in $G = (V, E)$ you have learnt in class is just one of many definitions of distances on a graph. Instead of the geodesic distance $d_G(u, v) := \min\{\ell(P) \mid P \text{ is a } u\text{-}v \text{ path in } G\}$ (a shortest $u\text{-}v$ path is also referred to as a $u\text{-}v$ geodesic), we will now look at the detour distance of a graph $D_G(u, v) := \max\{\ell(P) \mid P \text{ is a } u\text{-}v \text{ path in } G\}$ (a longest $u\text{-}v$ path is also referred to as a $u\text{-}v$ detour).

- a) Show that Bablu has a valid strategy by proving that (V, D_G) forms a metric space and D_G is thus a valid graph distance. [CO4]

[3 marks]

- b) Under this new definition of distance, show that analogous results hold: [CO1],[CO2]

- $\text{rad}(G) \leq \text{diam}(G) \leq 2\text{rad}(G)$.
- $|\varepsilon(u) - \varepsilon(v)| \leq D_G(u, v) \forall u, v \in V$.
- Q_n is still self-centered.

(Hint: Recall from tutorials that Q_n is Hamiltonian, i.e., there exists a cycle containing all vertices. You can use this fact directly.)

[1.5+1.5+2=5 marks]

- c) Show that Birendra can counter Bablu's strategy, by showing that there exists a connected graph G for each $|G| \geq 2$, such that $d_G(u, v) = D_G(u, v) \forall u, v \in V$. [CO1]

[2 marks]

5. The topic we spent the longest time on in class, was Connectivity. Let's test how much you remember of Connectivity, after it was asked in Quiz 2:

- a) Does there exist a disconnected graph with an Eulerian tour? Either prove that they do not exist, or give an example. [CO3]
- b) Show that any connected Eulerian graph is 2-edge-connected. [CO3]
- c) Prove or Disprove: If P is a u - v path in a 2-connected graph G , then G contains another u - v path P' , which is independent from P . [CO3]

(Hint: Note that Menger's only says that there should exist a pair of independent $u - v$ paths. So read the question carefully.)

- d) If G is a graph with $|V(G)| = n \geq k + 1$ and $\delta(G) \geq \frac{n + k - 2}{2}$, then prove that G is k -connected and has diameter at most 2. [CO3], [CO4]

(Hint: Can you count the number of common neighbours of a pair of vertices?)

- e) A question straight from the textbook: Show that every k connected graph ($k \geq 2$) with at least $2k$ vertices contains a cycle of length at least $2k$. [CO3], [CO4]

(Hint: First show that the graph always has a cycle. Then, consider the maximum cycle, and let it be of length less than $2k$. Let v be a vertex outside this cycle. Can you show that a fan from v to this cycle must contain a pair of adjacent vertices of this cycle? What does this imply?)

[1.5+1.5+2+4+6=15 marks]
